

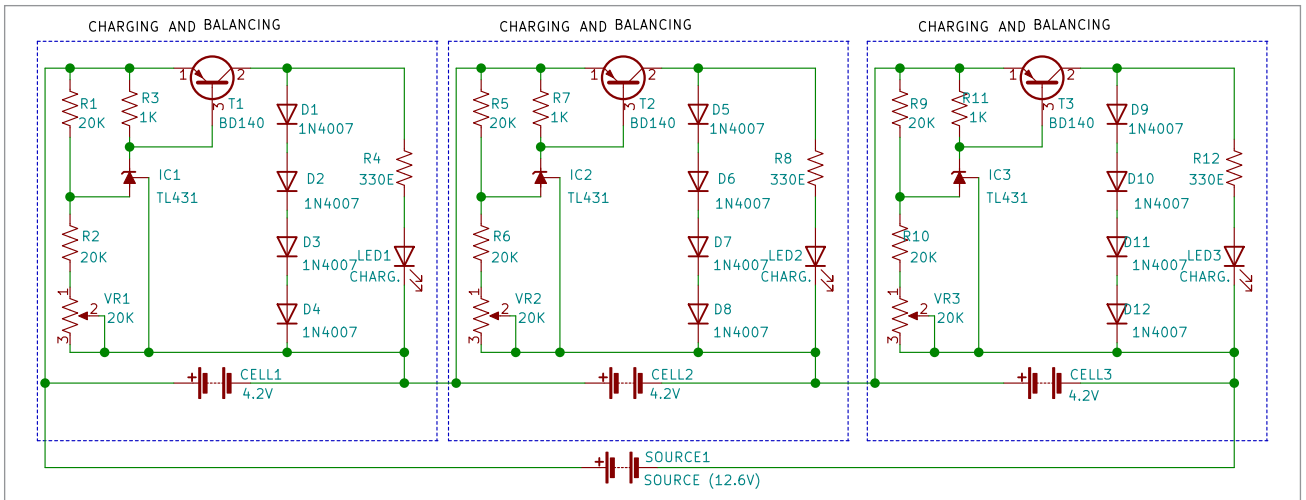


Fig. 5: Charging and balancing circuit for 3-cell Li-ion battery

minimal. The adjust pin ADJ of the IC is connected to the Vout with resistor R2 in between and to the ground through resistor R1.

To improve the transient response of the output, capacitor C2 is connected across Vout pin of the IC and the ground. The output voltage of IC1 can be adjusted as per following relationship:

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1} \right) + I_{adj} R_2$$

Since the value of I_{adj} is very small, it will have little effect on the output voltage. For a single cell Li-ion charger, the value of resistor R1 and R2 would be 3.3k and 1.4k, respectively. The values of R1 and R2 need to be changed for achieving different voltages for different cell chemistries or different battery configurations.

Construction and testing

An actual-size, single-side PCB for the charger circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Connect the input across J1 and the battery under charge (BUC) at BATT.1.

For setting the threshold voltage for the adjustable shunt regulator, in place of the cell connect a regulated power supply. Keep the output volt-

age the same as the threshold voltage you want to keep as cutoff voltage for the cell. Now, turn pot VR1 until LED1 starts glowing. This is the cutoff voltage at which the circuit will bypass the cell and current will start flowing through the series of diodes.

You can use the same circuit for charging different cell chemistries, such as lithium ion phosphate (LFP), lithium nickel manganese cobalt oxide (NMC), or even lithium-ion polymer (Li-Po) batteries, by setting the cutoff voltage for the adjustable shunt regulator TL431.

Charging more than one cell

To charge more than a single cell, which are connected in series, we need to replicate the charging and balancing circuit in series. The connections for a 3-cell charger are shown in Fig. 5.

From Fig. 5 it is clear that the charging and balancing circuits are connected in series with each other and the whole series is connected in parallel to a power source. The cells that need to be charged are connected individually to a charging circuit each.

Since the cells may take different times to charge and discharge, the voltage of each cell in the battery pack may not be the same. Hence, when one of the cells is charged, the

PARTS LIST

Semiconductors:

- IC1, IC2 - LM317 adjustable voltage regulator
- IC3 - TL431 shunt regulator IC
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 3.3-kilo-ohm
- R2 - 1.4-kilo-ohm
- R3 - 2-ohm, 1-watt
- R4, R5 - 20-kilo-ohm
- R6 - 1-kilo-ohm
- R7 - 330-ohm
- VR1 - 20-kilo-ohm pot

Capacitors:

- C1, C2 - 100 μ F, 25V electrolytic

Miscellaneous:

- J1 - DC jack/2-pin terminal connector
- BATT.1 - 2-pin terminal connector
- Breadboard
- Battery to be charged

threshold voltage of TL431 in that specific circuit is reached and the cell in that circuit is bypassed, so the current flows through the four diodes in series and the indicator LED turns on. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-constructing-li-ion-charger>

The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.